WORKSHEET

Annuity problems

A financial solver is required to complete this worksheet.

1 Martha invests $200 000 at 8.9% p.a. compounded yearly. Interest is paid at the end of each year and then Martha withdraws $24 000. The remaining money is invested for another year.

a Write a recurrence relation of the form $A_{n+1} = rA_n + d$, where $A_1 = a$ that gives the value of the investment A_n at the start of the nth year.

b How much money will remain in the investment after 4 years?

c How long will Martha's annuity last? Answer to the nearest year.

2 Druvham invests $30 000 in an annuity that pays interest at 9.75% p.a. compounded annually. Each year, after the interest is added to the loan, he withdraws $3000 which he spends on a holiday. The value of his investment, A_n, at the start of quarter n is given by a recurrence relation of the form:

$$A_{n+1} = rA_n + d, \text{ where } A_1 = a.$$

a Find the values of r, d and a.

b How long will this annuity last? Answer to the nearest year.

3 Hamza invests $150 000 in an annuity offering 7% interest per annum, compounded quarterly. Each quarter, Hamza receives a payment of $4000. The value of his investment, A_n, at the start of quarter n is given by a recurrence relation of the form $A_{n+1} = rA_n + d$, where $A_1 = a$.

a Find the values of r, d and a.

b How long will this annuity last?

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4 Jun inherits $90 000 and decides to invest it in an annuity. The annuity pays 12.5% interest per annum, compounded monthly and Jun receives a monthly payment of $1500.

a Write a recurrence relation of the form $A_{n+1} = rA_n + d$, where $A_1 = a$ that gives the value of the investment A_n at the start of the nth month.

b How much money will remain in Jun's annuity after 2 years?

c To the nearest month, how long will the annuity last?

5 On her 25th birthday, Georgia inherits an annuity of $1 000 000 that earns 0.5% interest each month. She is allowed to decide on the size of the monthly payment she will receive from the annuity.

a What should be the size of the payment if the value of the annuity is not to fall below $1 000 000?

b If Georgia sets the size of the payment to $7500, find the value of her annuity after 6 years.

6 On his retirement, Hamish invests his superannuation in an annuity. He invests $500 000 at 6.5% p.a., compounded monthly. Each month, he is to receive a payment. Hamish would like this investment to last for 25 years. What should be the size of the monthly payment?

7 Arabella establishes an annuity to support a yearly shopping spree. She invests $50 000 at 8.75% p.a. compounded annually and withdraws a payment once each year, after the interest has been paid. How much can she afford to spend from the annuity each year if:

a the annuity is to last 10 years

b the annuity is to last 20 years

c the annuity is to last 30 years?

8 Mary-Ellen has $75 000 that she invests in an annuity. Interest is paid at a rate of 8% p.a., compounded monthly. Mary-Ellen receives a monthly payment of $1000.

a How much will be left in the annuity after 7 years?

b How much interest has been earned in this time?

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9 Jem wins $315 000. He invests it at 10.5% p.a., compounded fortnightly, and receives a payment of $1800 per fortnight, after the interest is paid into the investment.

a How long will the investment last? Answer to the nearest fortnight.

b By how much should Jem reduce his fortnightly payment in order to make the investment last 10 years longer?

10 Yubin and her twin sister Jing win $120 000 that they share evenly, but invest differently. Yubin invests her money at 7.15% p.a., compounded monthly, and receives a monthly payment of $700. Jing invests her money at 7.2% p.a., compounded fortnightly and receives a fortnightly payment of $350.

a Whose investment will last longer, and by how much? Round your answers to the nearest month or fortnight.

b Who will receive more money from their investment over the entire term of the annuity?

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Answers

1 **a** $A_{n+1} = 1.089A_n - 24\,000$ where $A_1 = 200\,000$

b $ 171 688.40

c 16 years

2 **a** $r = 1.0975$, $d = -3000$, $a = 30\,000$

b 40 years

3 **a** $r = 1.0175$, $d = -4000$, $a = 150\,000$

b 62 quarters or 15.5 years

4 **a** $A_{n+1} = 1.0104167A_n - 1500$ where $A_1 = 90\,000$

b $74 714.59

c 95 months or 7 years and 11 months

5 **a** $5000

b $783 977.86

6 $3376.04

7 **a** $7705.48

b $5380.09

c $4759.29

8 **a** $18 943.35

b $27 943.35

9 **a** 304 fortnights or 11 years and 18 fortnights

b $381.82

10 **a** Yubin's, by 1 year and 1 fortnight

b Yubin

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